uCNC RP2350B Motherboard Specification - PRE-FINAL

Generated review documents:

- Combined board review
- Front artwork and assembly layer
- Mirrored rear artwork
- Current top 3D render
- Current bottom 3D render

Board summary

Item	Specification
Product	LeanCam NCC / uCNC RP2350B Motherboard
Status	PRE-FINAL; manufacturing review outstanding
Controller	Socketed WeAct Studio RP2350B Core Board
PCB outline	160 mm x 120 mm
PCB construction	Two copper layers, 1.6 mm nominal FR-4
Mounting	Four M3 clearance holes
Assembly	Main population on F.Cu; J22 is also on F.Cu
Main input	12-24 V DC machine supply
Main logic rail	LMR36510 5 V, 1 A synchronous buck
Peripheral logic	AMS1117-3.3, `3V3`
Wi-Fi supply	Dedicated AMS1117-3.3, `WIFI_3V3`
Analog-output rail	L78L12, `AO_12V`
PLC supply	PLC24+ PLC- through J33/J45
Backlight rail	LMR36510 isolated 24 V to `BL_12V`

The board is a machine-control carrier, not a safety-rated controller. Hazardous output power must be removed through an independent hard-wired safety circuit.

Digital I/O

Function	Capacity	Implementation
Isolated PLC inputs	16	U18-U21 TLP291-4, RN32-RN35 pull-ups
Dedicated E-stop	1	U35 TLP291 and R74 pull-up
Servo ZERO inputs	7	Direct 3.3 V logic into U6/U7, 10 k pull-ups
Servo ERROR inputs	7	Direct 3.3 V logic into U6/U7, 10 k pull-ups
Spare register inputs	2	U7 D2/D3 on J55 with 10 k pull-ups
General field outputs	16	U61-U64 optocouplers into U26/U27 TBD62083AFG arrays
Motion outputs	7 axes	A0-A6 STEP/DIR, with grouped `EN3` and `EN4` enables
Encoder inputs	3	A/B/Z, 5 V to 3.3 V through U28
Analog inputs	4	0-10 V resistor dividers into RP2350 ADC pins

Analog outputs	4	TLV2374 and two-stage RC filters
Storage	2 interfaces	J36 breakout or on-board J54 socket

The input chain runs from U6 to U7 to U8 to U9 and returns on GPIO23. U8/U9 carry PLC inputs 1-16, while U6/U7 carry the seven paired servo ZERO/ERROR signals. RN45-RN48 provide a 10 k pull-up on every U6/U7 data input. Two unused U7 data inputs are available on J55 with their pull-ups setting the default high state; U7 pin 15 remains tied low because it is `CLK INH`. The output chain runs from GPIO20 through U13, U12, U11 and U10. GPIO21 is the shared shift clock and GPIO22 is the 74AHCT595 latch.

The exact register-bit assignments are in Input and output bit map.

Field I/O

J33 and J45 are parallel two-pin access points for PLC 24+ and PLC -. The PLC domain supplies the contact loops and output side and has no connection to motherboard GND.

Each PLC input contains one connector-side indicator LED, 2.2 k field resistor, reverse diode and TLP291-4 channel. Each four-channel group has one RN32-RN35 4 x 10 k logic-side pull-up network. The dedicated E-stop has its own indicator, series resistor, reverse diode and TLP291 channel.

The sixteen general outputs are generated by four 4 x 1 k network arrays, optically isolated through U61-U64 and switched by U26/U27. The output arrays are low-side switches; external loads must be supplied from the isolated field rail and kept within the TBD62083AFG package limits.

E-stop is an input to the controller only. It does not locally interrupt motion or general-output power.

Motion and encoders

J10, J11, J28, J29, J47, J46 and J30 expose A0-A6 from top to bottom. Each connector also carries the applicable grouped enable:

- `EN3` is shared by A0-A2.
- `EN4` is shared by A3-A6.

The visible PCB labels follow the rotated connector geometry and read `DIRn STPn ENx`; electrically, pin 1 remains the enable, pin 2 STEP and pin 3 DIR.

All 275 component reference designators are visible on silkscreen and placed outside exposed pads, vias and existing silkscreen features. Each reference is first attempted at the component center; small passives whose 0.8 mm reference cannot fit legally inside the body receive the nearest clear adjacent position. All 309 visible footprint reference/value fields use a uniform 0.8 mm text size and 0.12 mm stroke. Functional LED values such as `D01–D16` remain documentation-only on `F.Fab`.

The 5 V SN74AHCT595 outputs drive two MC74AC541DWR2G wide-body SOIC-20 buffers in the 7.5 x 12.8 mm package. The buffers provide the 5 V STEP, DIR and enable outputs.

J40, J41, J42, J43, J50, J49 and J48 are the paired inputs for A0-A6. Each is 3-pin: `GND`, `ERR`, `Z`. J13-J15 are 5 V encoder connectors. U28 converts the 5 V A/B/Z push-pull signals to 3.3 V logic; open-collector encoders require external pull-ups.

Display, storage and expansion

J22 is a 2 x 10, 2.00 mm LCD interface on F.Cu. It carries four HSTX differential channels through RN26/RN27 4 x 330 ohm, 4 x 0603 arrays. Pins 1-6 supply 3.3 V and ground, pins 7-16 carry D0, D1, D2 and clock, and pins 17-20 are unconnected.

J2 carries isolated `BL_12V` and `PLC_0V` for the backlight. PWM switching, current limiting, any CCFL inverter and high-current LED drive remain off board.

J36 accepts a passive Open-Smart 3.3 V TF Card Breakout. J54 is a SOFNG TF-15x15 `C111196` microSD socket wired in parallel. Fit either card interface for use and never insert a card into both simultaneously. RN43 provides pull-ups and RN44 provides 22 ohm series resistors; JP5 enables the card-detect switch.

J51 exposes GPIO4-GPIO8, `3V3` and GND on a 1 x 7, 2.54 mm socket. J20 accepts an ESP-01 or ESP-01S on the separately regulated `WIFI_3V3` rail.

Power

The main logic path is protected by D1 and D2. U1 converts the protected input to `5V`; the WeAct module is supplied from that rail and uses its onboard 3.3 V regulator. U2 supplies the motherboard peripheral rail, and U34 supplies the optional ESP module independently.

U3 converts the isolated 24 V field input to `BL_12V` and is protected by D3. U37 creates the low-current `AO_12V` rail for U38 and the four 0-10 V outputs. All regulator loops, bulk capacitors, bypass capacitors and feedback networks are documented in Decoupling map.

The main switching return, peripheral logic and analog return share one GND plane. `PLC_0V` remains a separate isolated domain.

Connector inventory

Ref	Function
J1	PCB label order `GND 24+`; pin 1 is `24+`, pin 2 is `GND`
J2	Isolated backlight `BL_12V` / `PLC_0V`
J5	Isolated E-stop loop
J7, J8	PLC input signals 1-16
J10, J11, J28, J29, J47, J46, J30	A0-A6 STEP/DIR outputs
J12, J32	General outputs D01-D16
J33, J45	Parallel PLC24+ PLC-
J13-J15	Three encoder interfaces
J40, J41, J42, J43, J50, J49, J48	A0-A6 paired status inputs
J18	5-pin I2C interface
J20	ESP-01
J22	2 x 10 LCD LVDS interface
J24	Four 0-10 V analog inputs plus ground
J25	Four 0-10 V analog outputs plus ground
J36	1 x 9 TF-card breakout header
J51	GPIO expansion socket
J54	On-board SOFNG TF-15x15 microSD socket

Complete pinouts are in Connector map. Net and connector abbreviations are in Signal naming.

Verification

Current KiCad 10 checks:

- Schematic: 277 components and 321 nets, with no connectivity failures.
- PCB: 271 footprints, 2,227 routed track segments and 225 vias.
- DRC: zero unconnected items and two courtyard overlaps: the pre-existing D1/J1 corner and C107/C108 after the requested C108 package enlargement.
- The 108 reference/value-field warnings were cleared by moving 57 DRC-confirmed fields to nearby clear silkscreen positions.
- A further 130 references were promoted from `F.Fab`; all 275 component references are now visible without introducing new DRC warnings.

- Remaining warnings are two small LVDS row legends below the configured text height and two J54 outline segments over GND pads.
- ERC: no electrical errors; 35 library-symbol cache warnings.

The board is electrically complete. The remaining warnings concern the library-provided microSD outline and the dense LVDS row legends; they do not concern component reference/value readability.

